Tutorial - 4

An email service provider wants to classify emails as spam or not spam, using logistic regression. Data from 10 emails has been collected.

Input features:

no. of suspicious words
no. of external links
Presence of attachments (0 = NO, 1 = yes)

Target varibles (1 = spam, 0 = not spam)

suspicious words (x_1)	external links (x_2)	Attachment (x_3)	Spam (y)
5	2	1	1
1	0	0	0
3	1	0	1
0	0	0	0
6	3	1	1

1. Identify the independent and dependent varibles.

Independent varibles

x_1 = suspicious words
x_2 = External links
x_3 = Attachment
$x = (x_1, x_2, x_3)$

Dependent varibles

y = spam (1 = spam, 0 = no spam)

2. write the logistic regression model equation.

logistic regression model

$$\boxed{z = w_0 + w_1 x_1 + w_2 x_2 + w_3 x_3}$$

Sigmoid function

$$\boxed{\sigma(z) = \frac{1}{1+e^{-z}}} \quad \boxed{\hat{y} = \sigma(z)}$$

3. Cost function:

$$\boxed{J = -\frac{1}{N} \sum_{i=1}^{N} [y \log(\hat{y}) + (1-y) \log(1-\hat{y})]}$$

x_1	x_2	x_3	y	$\hat{y}$	$y - \hat{y}$
5	2	1	1	0.5	0.5
1	0	0	0	0.5	-0.5
3	1	0	1	0.5	0.5
0	0	0	0	0.5	

$w_0 = 0 \quad w_1 = w_2 = w_3 = 0$

$z_1 = 5(0) + 2(0) + (1)1 + 1 = 1 \qquad \hat{y} = \sigma(z_1) = 0.73$

$z_2 = 1 \qquad \hat{y} = 0.73 \qquad \hat{y} = 1$

$z_3 = 1 \qquad \hat{y} = 0.73 \qquad \hat{y} = 1$

$z_4 = 1 \qquad \hat{y} = 0.73 \qquad \hat{y} = 1$

$z_5 = 1 \qquad \hat{y} = 0.73 \qquad \hat{y} = 1$

Accuracy $= \frac{3}{5} = 60\%$

$$J = \frac{1}{N} \sum_{i=1}^{n} \left[y_i \log(\hat{y}_i) + (1-y_i) \log(1-\hat{y}_i) \right]$$

$$= \frac{1}{5} \Big[(1) \log(0.73) + (1-1) \log(1-0.73) + (0) \log(0.73) + (1-1) \log(1-0.73)$$
$$(1) \log(0.73) + (1-1) \log(1-0.73) + (0) \log(0.73) + (1-0) \log(1-0.73)$$
$$(1) \log(0.73) + (1-1) \log(1-0.73) \Big]$$

$$= -0.713 \rightarrow [-0.713] = \underline{0.713}$$

<u>Updating weights and bias:</u>

$$\boxed{w_0^{new} = w_0^{old} + \eta \frac{1}{N} \sum_{i=1}^{n} (y_i - \hat{y}_i)}$$

$$= 1 + 1\left(\frac{1}{5}\right)\left[(1-0.73) + (0-0.73) + (1-0.73) + (0-0.73) + (1-0.73)\right]$$

$$= 1 + \left[\frac{-0.65}{5}\right] \rightarrow 1 - 0.13 = \underline{0.87}$$

$$w_1^{new} = w_1^{old} + \eta \left[\frac{1}{N} \sum_{i=1}^{n} (y_i - \hat{y}_i) x_i \right]$$

$$w_1 = 0 + 1 \times \frac{1}{5}\left[(0.27)(5) + (-0.73 \times 1) + (0.27 \times 3) + (-0.73 \times 0) + (0.27 \times 6)\right]$$
$$= 0.302$$

$$w_2 = 0 + 1 \times \frac{1}{5}\left[(0.27)(2) + (-0.73 \times 0) + (0.27 \times 1) + (-0.73 \times 0) + (0.27 \times 3)\right] = \underline{0.32}$$

$$w_3 = 0 + 1 \times \frac{1}{5}\left[(0.27)(1) + (-0.71)(0) + (0.27)(0) + (-0.73)(0) + (0.27)(1)\right]$$

$$= \underline{0.108}$$